

# Buoyancy fluxes of sea-ice leads in the Arctic

Josué Martínez-Moreno<sup>1</sup>, Camille Lique<sup>1</sup>, Claude Talandier<sup>1</sup>, someone else?

<sup>1</sup>Laboratoire d’Océanographie Physique et Spatiale (LOPS), University of Brest/IFREMER/IRD/CNRS,  
Brest, France

## Key Points:

- Buoyancy fluxes in Linear Kinematic Features vary seasonally
- Buoyancy fluxes in Linear Kinematic Features can be 20-30% larger than in the ice-pack.
- The presence of Linear Kinematic Features has the potential transform water masses in the ocean surface.

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Corresponding author: Josué Martínez-Moreno, [josue.martinez.moreno@ifremer.fr](mailto:josue.martinez.moreno@ifremer.fr)

## Abstract

The Arctic sea-ice, in particular the ice-pack, acts as an insulator between the atmosphere and the ocean below. Linear kinematic features (LKF), or leads, common in Arctic sea ice serves as openings that allow exchanges between ocean and atmosphere, for example: heat, salt, moisture, and gases. Therefore, leads can influence heat exchanges, ice melt or growth, ocean dynamics, and the atmospheric boundary layer. The scale of leads varies from a few meters to kilometers in width, and from few kilometers up to hundreds of kilometers in length. Their formation is driven by the winds and ocean currents. Existing literature has suggested that leads play a significant role in the heat budget of the surface of the ocean and likely impact water mass transformation within the Arctic. However, due to the lack of observations and high resolution models resolving LKFs, an estimate of the magnitude of the impact in the ocean has been elusive. Here, we explore the magnitude and impact in the ocean surface caused by the LKFs using SEDNA, a high resolution pan-Arctic hindcast ( $1/60^\circ$ ; NEMO + SI3 configuration). We show that despite the LKF only account for a small area of the sea-ice cover ( $\sim 5\%$ ), they contribute up to 20-30% more buoyancy than the ice-pack at the surface. Furthermore, of the total surface water mass transformation in the Arctic ice cover, LKFs explain approximately 10% of it. LKFs make a significant contribution despite their small area coverage; however, the presented estimate shows a large local impact but to a lesser degree in the Arctic Ocean than previously hypothesized.

## 1 Introduction

The Arctic sea-ice plays a crucial role in regulating Earth's climate by serving as a natural insulator between the atmosphere and the ocean below (Wettlaufer et al., 1997; Untersteiner, 1961). This insulator property acts as a thermal barrier, limiting heat loss from the ocean to the atmosphere and minimizing evaporation, thereby it maintains the stability of the polar environment (Ruffieux et al., 1995). An ubiquitous feature of the Arctic sea ice is the formation and persistence of linear kinematic features (LKF), commonly known as leads. These features are elongated openings of the sea ice cover that expose the ocean surface to the atmosphere (Lüpkes et al., 2008). Sea ice leads serve as conduits for heat, moisture, and gas exchanges between the ocean and the atmosphere (Michaelis & Lüpkes, 2022; Andreas et al., 2002). Depending on the season and the atmospheric forcing, this direct contact between the ocean water and atmosphere can re-

sult in important heat transfers at the ocean-atmosphere interphase (Marcq & Weiss, 2012), localized ice melt or growth, and modify the ocean dynamics (Bourgault et al., 2020a).

Sea ice leads vary in scale from a few meters to kilometers in width, and span distances of a few kilometers up to hundreds of kilometers (Tschudi et al., 1998). Their formation mechanisms are diverse, ranging from mechanical to thermodynamic processes (Rampal et al., 2016; Hutchings et al., 2005). Mechanically, winds and ocean currents cause the sea ice to drift, deform, and fracture (Richter-Menge et al., 2002). While winds are thought to be the main forcing in the sea ice deformation, due to their turbulent nature and their varying temporal and spatial variability, ocean drag at the bottom of the sea ice balances the sea ice wind induced velocities.

later one, wind forcing induces stress and strain

-induced stress in the sea ice results in ice fracturing, which can eventually evolve into leads. (Linow & Dierking, 2017; Hutter et al., 2018).

Ocean currents contribute by generating lateral forces on the ice, stretching it and causing fractures along zones of vulnerability. Particularly influential are regions of strong currents and topography features (Willmes et al., 2023). Alongside mechanical forces, thermodynamic processes also shape sea ice leads as temperature differences between the ocean waters and the atmosphere can induce localized melting and freezing, influencing the overall configuration of the sea ice leads.

Sea ice leads are crucial since they act as gateways between the atmosphere and the underlying ocean. These opening of the ice cover results in intensified heat and salt exchange between the ocean and atmosphere, for example, in winter, ice growth in the lead results in brine rejection in the ocean surface, and result in convective processes in both the ocean and atmosphere (e.g., Lüpkes et al., 2008; McPhee et al., 2005). As the ocean surface interact with the atmosphere above the leads, the surface waters are transformed, since changes in the surface distribution of temperature and salinity modify the ocean density and thus the surface water masses are transformation (Rabe et al., 2011; Wallin et al. XXXX). Existing literature have suggested that leads play a significant role in the heat budget of the surface of the ocean and likely impact water mass transformation within the sea-ice of the Arctic (McPhee et al., 2005; Wettlaufer et al., 1997). How-

ever, the magnitude and influence of the LKFs compared to the surrounding ice pack in the Arctic basin is yet to be quantified.

The presence of sea ice leads has a non-local impact, since they can alter the stability of the surrounding ice pack (Bourgault et al., 2020b). The exposure of ocean water to the Arctic atmosphere leads to localized ice growth, creating new ice and modifying the ice pack’s overall structure (Rampal et al., 2009). Additionally, leads contribute to the dispersion and fragmentation of the ice cover, affecting its overall extent and concentration (Tschudi et al., 2016). The increased surface area of leads relative to the surrounding ice pack also influences radiative heat exchange, further impacting the rate of ice melt and overall ice-albedo feedback (Lindsay and Schweiger, 2015). Investigating the influence of leads on the ice pack’s dynamics is crucial for understanding the sea ice patterns in the Arctic.

## 2 Methods

### Model description

The study employs the NEMO numerical platform (release 4.0.5) to establish a  $1/60^\circ$  pan-Arctic ocean-sea ice configuration named SEDNA. The ocean core (NEMO-OCE) solves the primitive equations (PE) using finite differences on an Arakawa C-grid mesh, while the ocean is coupled to the SI3 sea-ice component (NEMO-SI3). The pan-Arctic domain covers the Bering Strait on the Pacific side and extends southward to  $56^\circ\text{N}$  in the Subpolar North Atlantic gyre and to the Baltic Sea entrance. The ocean model incorporates a linear free surface, utilizes a 3rd order flux form scheme for momentum advection, and employs a 4th order Flux Corrected Transport (FCT) for tracer advection. Additionally, the sea-ice representation includes a 5-category model with the Elasto-Visco-Plastic (EVP) rheology activated and a land-fast capability to better simulate sea-ice behavior in specific areas.

The NCAR bulk formula from Large & Yeager (2009) to calculate air-sea fluxes and uses hourly ERA5 atmospheric data for near-surface variables. The three lateral boundaries are constrained with daily GLORYS12V1 Re-analysis data, and monthly freshwater fluxes from river discharges are combined with Greenland land ice melt data to supply SEDNA. The simulation extends over seven years (2009 to 2015) and starts from rest, utilizing initial conditions based on the World Ocean Atlas 2009 temperature/salinity



and mean January 2009 ice state from PIOMAS re-analysis. Preliminary assessment against available observations and re-analyses data shows that the simulation reasonably captures key ocean and sea-ice characteristics.

### LKF identification

The LKF from SEDNA were identified by using the algorithm proposed by Hutter et al. (2019). This method uses the total deformation rate defined as:

$$T_d = \left[ \underbrace{\left( \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right)^2}_{Divergence} + \underbrace{\left( \frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \right)^2 + \left( \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right)^2}_{Shear} \right]^{1/2}, \quad (1)$$

where  $u$  and  $v$  are the sea ice velocities. The deformation rate varies spatially depending on the background deformation and the sea ice properties (), where the largest deformation rates occur along the boundaries of ice flows where the LKF are found. Therefore, to identify the LKFs it is required to identify the local maxima using edge detection (DoG filter) in the deformation rate field. After the edge detection, using binary map we obtain a mask with the original width of identified LKF. This mask is then used to compute the statistics presented in this study. After this step, Hutter et al. (2019) reduce the LKFs width to a single pixel to then track the LKF over time. Since our study focus on the ocean-atmosphere interactions, we only use the identification algorithm to produce a mask of LKFs during 2014 for the Arctic, where we retain the original width computed from the deformation rate. This mask is then used to quantify and contrast the impact of LKFs compared to the ice-pack.

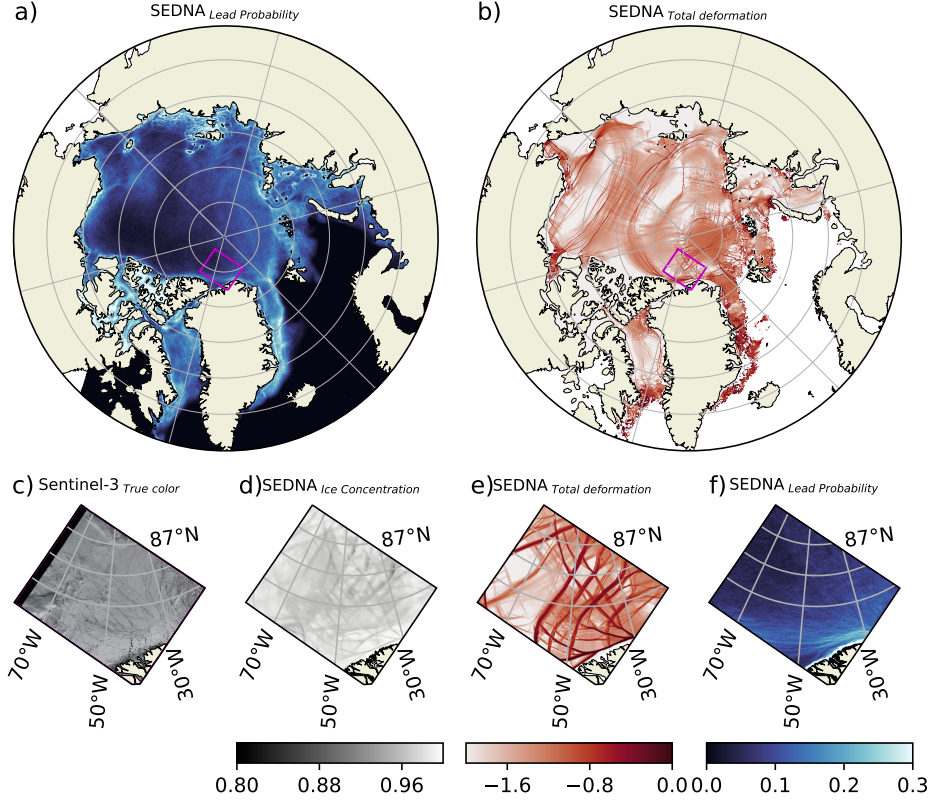
### Buoyancy flux and water mass transformation

- Describe MLD definition
- Describe buoyancy flux and decomposition between thermal and haline components.

## 3 Results

### Figures:

1. Satellite and model image of LKF
2. Buoyancy flux on 2 different snapshots and LKF mask (Total deformation panel?).



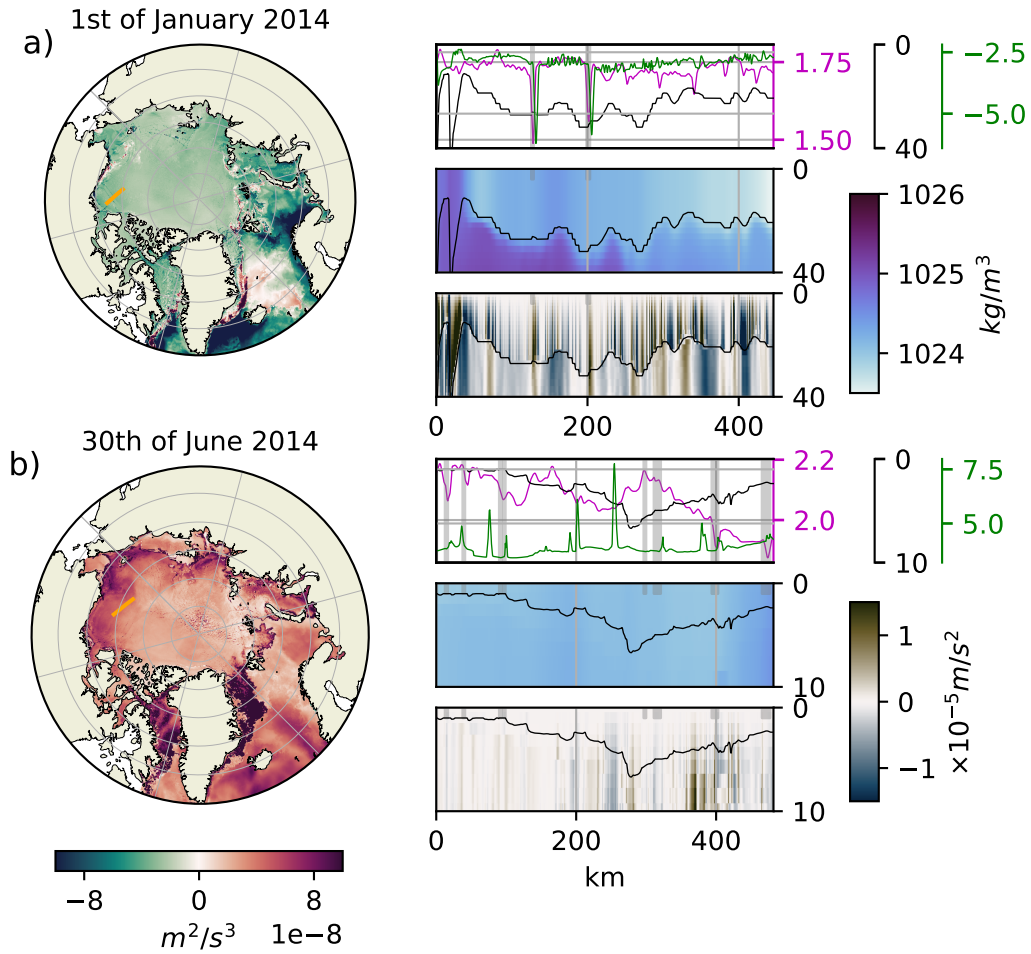
**Figure 1.** Linear kinematic features from observations and high resolution model. a) True color from Sentinel 3 on the 12th of April 2022. b) Zoom of magenta area in panel a) showing the linear kinematic features on the 12th of April 2022. c) Mean ice concentration from SEDNA (800m resolution) on the 21st of April 2014. d) Zoom of magenta area in panel d) showing the linear kinematic features on the 21st of April 2014.

3. Time-series of area cover + buoyancy fluxes. (Merge the 0-80
4. Water mass transformation of the Pack Ice and the LKF.
5. Density composite for summer and winter? (Potential to affect interior)

### 3.1 Observation vs Models

#### Figure 1

- Describe spatial pattern of features, for example in satellite we observe many floes of different scales that the model can not reproduce
- Describe that the model is a mean state over a day, thus it is likely smoother.



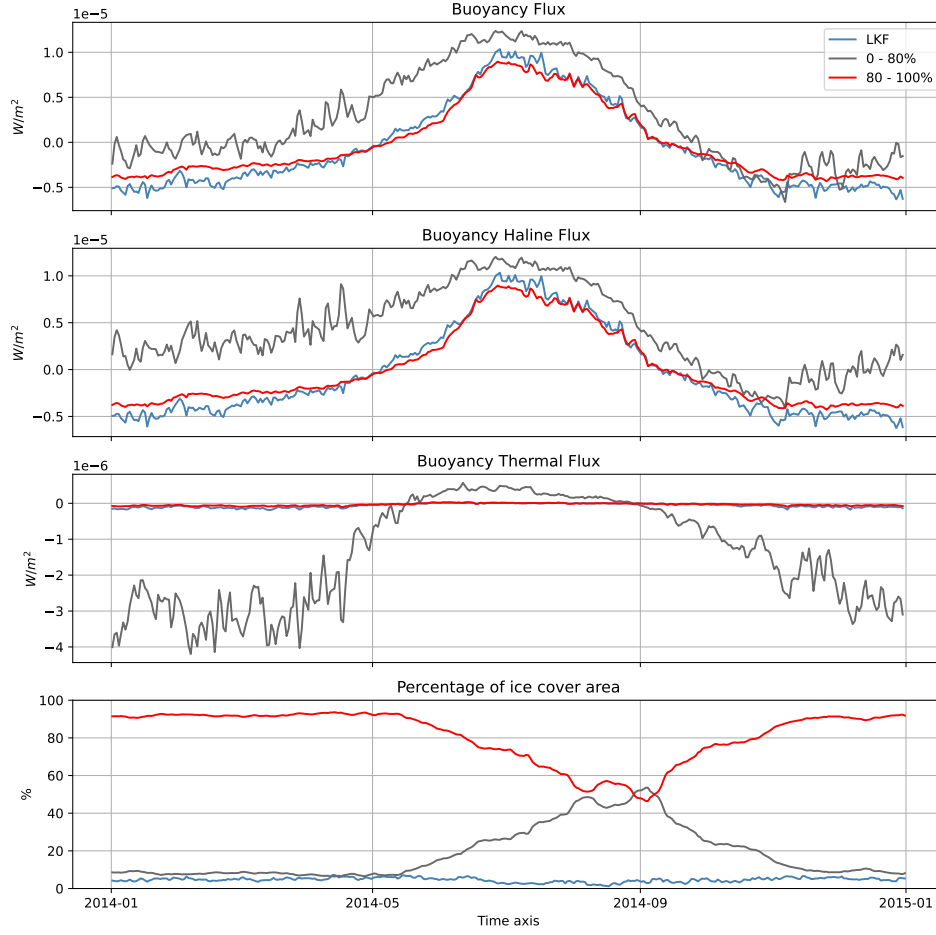
**Figure 2.** Buoyancy flux and Ekman Velocity for two snapshots during winter and summer.  
a) Buoyancy flux in Winter the 1st of January 2014. b) Buoyancy flux in Summer the 15th of June 2014. c) Ekman velocity in winter the 1st of January 2014. d) Buoyancy flux in summer the 15th of June 2014

- Highlight that model still reproduces features similar to those observed in the model.
- The length-scales also appear to match (Should we do something to quantify this?)
- Add statistics of the LKF compare to other studies?

### 3.2 Impact of LKF in the ocean buoyancy

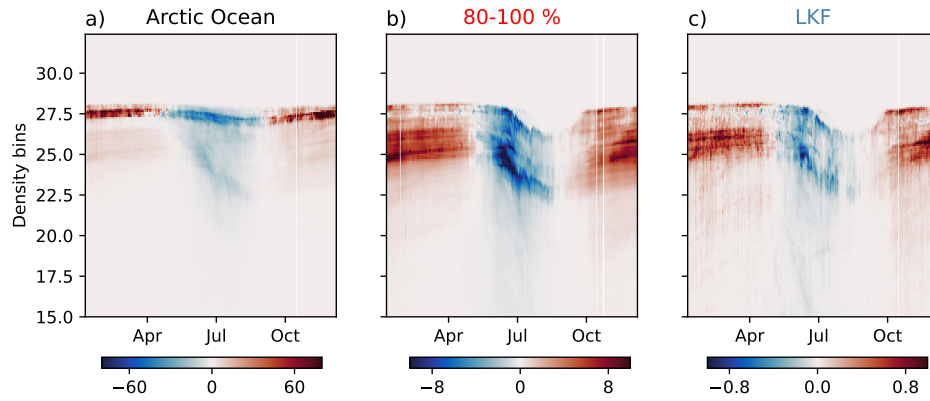
#### Figure 2

- Describe the sign of the buoyancy flux (Negave = denser water).



**Figure 3.** Buoyancy flux and Ekman velocities normalized by the ice cover area of linear kinematic features, transitional ice zone (ice concentration 0-80%), and pack ice (ice concentration >80%).

- Describe that we can see important differences in the BF between the open ocean and the packed ice.
- Note some of the LKF features observed in the buoyancy flux, for example large negative flux in the East Siverian Sea or Beaufort sea and the positive flux in the center of the Arctic.
- Describe analogously the Ekman velocities, negative = Ekman Pumping = downwelling.
- In the regions where we see larger buoyancy flux, we also observe large Ekman velocities.
- Talk about how the Ekman velocity depends on the atmospheric forcing.



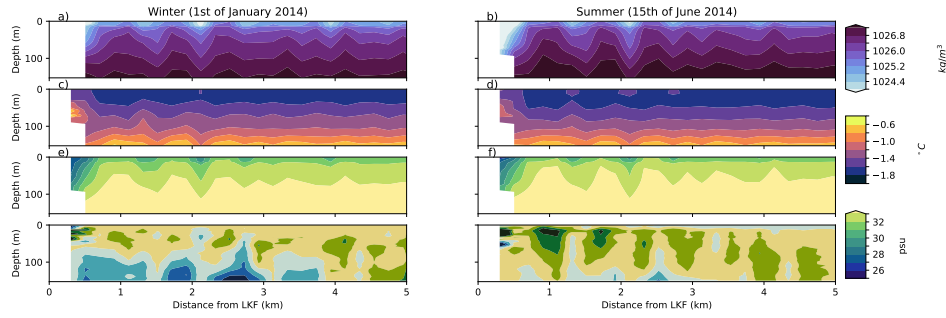
**Figure 4.** Hovmöller diagram of the water mass transformation (sv). Water mass transformation of for a) the Arctic ocean, b) pack ice, c) and linear kinematic features. Note the changes in the magnitude of the colorbar of each panel.

### Figure 3

- This figure will be updated
- TODO: Remove 0% concentration, merge 0-80% concentration.
- TODO: Add time series of Ekman Velocity. Remove the Thermal component of BF.
- Describe that the Buoyancy flux is dominated by a seasonal cycle with small high frequency variability.
- Important seasonal cycle, buoyant water in Summer, denser in Winter.
- Describe that area in LKF is really small  $< 5km$ .
- Fluxes are mostly dominated by haline component
- Fluxes in LKF are comparable to those in the Pack ice.
- Explain that despite the area being just a fraction, generally less than 10% of the ice-covered Arctic, the fluxes in LKF explain the fluxes of the packed ice and their magnitude is comparable to the marginal ice zone.

### Figure 4

- TODO: Add units below colorbar.
- Describe seasonal cycle of water mass transformation
- Highlight the change in the magnitude



**Figure 5.** Composite of density, temperature, and salinity normal to the identified linear kinematic features on the 1st of January and 15th of June 2014. a) Density in Winter the 1st of January 2014. b) Density in Summer the 15th of June 2014. c) Temperature in winter the 1st of January 2014. d) Temperature in summer the 15th of June 2014. e) Salinity in winter the 1st of January 2014. f) Salinity in summer the 15th of June 2014

- The LKF explain around 10% of the Pack Ice WMT.
- During summer the transformation is towards buoyant water and in winter the transformation is towards denser water.
- Describe shift of water mass transformation during summer.
- Is it possible to estimate the effect of water masses transformation based on the outcrop of density contours in the Arctic?

### 3.3 Impact of LKF in the ocean interior

#### Figure 5

- The composites shown important signature of buoyant and fresher water near adjacent to the LKF.
- The lense of buoyant water appears to vary with the season, but more info is needed to conclude.
- The average response near the LKF appears to be the same independent to the season but with an intensification in summer (melting in general in LKFs?), more needs to be done to conclude.
- We observe other lenses of buoyant and fresher water, likely linked to past LKF events.
- Tilting of isopycnals near the LKF?

- Warmer water near the LKF? connection between the intermediate water and the surface through LKF?
- What is the role of mixing here? since the water near the LKF is really buoyant?

Should we do a schematic (Fig 1) as in Michaelis, J. & Lüpkes, C. but for the ocean?

## 4 Conclusions

reference (Bouchat et al., 2022) to argue that it's ok to use this high resolution simulation since they seem to resolve properly the LKF.

## 5 Open Research

## Acknowledgments

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